SpectralBL Boundary Layer Analysis Report:

Reconstructing the Nocturnal Attractor Manifold

Automated Analytics Pipeline Engine
SpectralBL-Analytics Framework

June 15, 2026

Abstract

This report provides an automated, low-rank structural analysis of stable boundary layer (SBL) regimes using data from the CASES-99 and GABLS3 field campaigns. Classical gradient metrics, such as the gradient Richardson number (Ri_g), frequently fail to resolve non-stationary transitions, submesoscale gravity waves, and intermittent turbulent bursts under strongly stratified conditions. By projecting high-dimensional physical profile measurements onto a low-rank, continuous phase space via a p-FEM operator, we reconstruct the underlying attractor manifold. This framework provides an information-theoretic approach to diagnosing boundary layer complexity and mapping regime transitions where traditional surface-flux similarity theories break down.

1 Introduction and Physical Context

Traditional boundary layer parameterizations rely heavily on local gradient profiles and standard Monin-Obukhov Similarity Theory (MOST). However, in the very stable boundary layer (vSBL), the coupling between the surface layer and the upper air column is frequently severed by the development of a strong thermal inversion. The atmosphere enters a regime dominated by radiative cooling, low-level jet (LLJ) development, global intermittency, and submesoscale internal gravity waves.

In these conditions, local turbulence metrics such as Ri_g become unreliable diagnostics: once Ri_g exceeds the critical stability threshold of 0.25, classical surface-layer similarity theory predicts the collapse of continuous turbulence. In practice, however, the boundary layer does not simply switch off. Intermittent bursting, wave-mode excitation, and the residual decay of turbulent kinetic energy continue in ways that local gradient metrics are fundamentally blind to.

To track these non-local structural reconfigurations, the SpectralBL pipeline extracts continuous spatial pattern modes of the boundary layer state. The resulting low-rank coordinates (η_1, η_2, η_3) correspond directly to deep physical phenomena: bulk background layer evolution, shear-driven jet modulation, and vertical wave-like structural curvature. These coordinates remain dynamically active throughout stable and super-critical-Ri_g conditions, providing information about boundary layer behavior that gradient-based scalings cannot access.

2 Attractor Dynamics and Low-Dimensional Phase Space

2.1 Overview

The boundary layer is analyzed as a low-dimensional dynamical system through projection onto dominant singular vectors. The following section presents the reconstructed phase space trajectory for campaign `CASES-99` using the v0.1 baseline.

Campaign: CASES-99

Baseline Version: v0.1

Total Samples: 6538

Mean Entropy: $H_{\text{mean}} = 0.211$

2.2 3D Phase Space Projection

Figure 1 visualizes the three-dimensional trajectory (η_1, η_2, η_3). To highlight the true multi-regime orbital evolution, timesteps are connected sequentially by a continuous historical path line. The trajectory is colored using a high-contrast colormap driven by the **Attractor Spin** diagnostic $(\Omega(t))$, explicitly isolating the directionality of regime transitions.

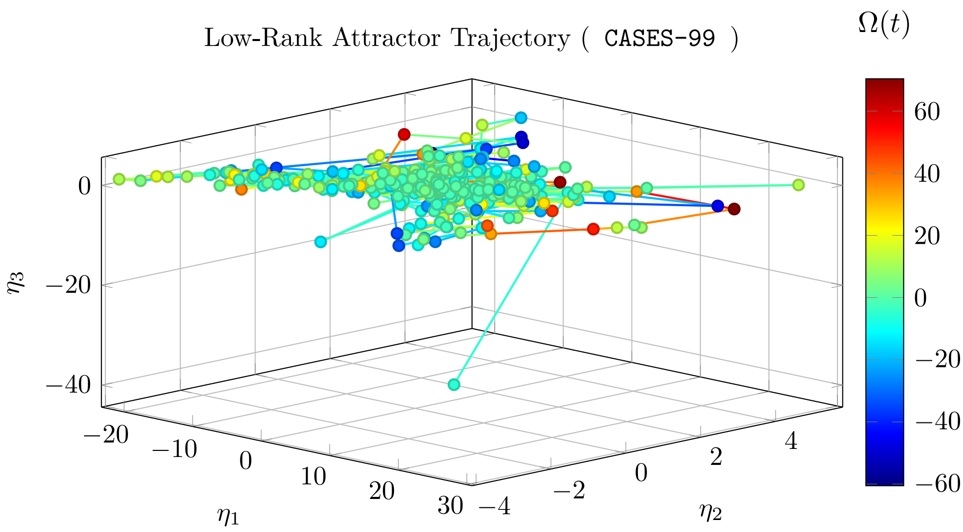

Figure 1: 3D attractor state orbit trajectory. Color variation tracks Attractor Spin (Ω), separating shear breakout acceleration (cool blue) from radiative collapse configurations (warm red).

2.3 Physical Coordinate Interpretation

This phase-space view acts as the compressed state engine of the atmospheric boundary layer (ABL), projecting high-dimensional tower profiles into three dominant, orthogonal coordinates:

- η_1 (**Bulk Background Mode**): Captures the mean inversion strength, nocturnal cooling depth, and overall boundary-layer depth evolution.
- η_2 (**Shear & LLJ Mode**): Captures shear production, alongside low-level jet (LLJ) speed and height modulation.
- η_3 (**Vertical Structural Curvature Mode**): Captures submesoscale gravity-wave excitation and the precursor structural signatures of intermittent turbulence events.

2.3.1 Geometric Reading Guide

- **Tight closed loops** indicate near-repeatable diurnal cycling (convective daytime mixing to stable nighttime recovery and back).
- **Broad, tangled trajectories** indicate non-stationary forcing, intermittency, or mesoscale disruption.
- **Color evolution by** $\Omega(t)$ isolates transition directionality: positive spin values (warm tones) indicate stable collapse paths; negative values (cool tones) mark turbulent regeneration breakouts.

2.4 Trajectory Metrics

The mean low-rank coefficients across all samples are:

$$\langle \eta_1 \rangle = 0.583 \tag{1}$$
$$\langle \eta_2 \rangle = 0.289 \tag{2}$$
$$\langle \eta_3 \rangle = -0.080 \tag{3}$$

The magnitude of the attractor vector $\|\boldsymbol{\eta}\|$ evolves continuously throughout the campaign, reflecting the transient structural adjustments driven by diurnal heating and nocturnal radiative cooling cycles.

Operationally, peaks in $\|\boldsymbol{\eta}\|$ often coincide with rapid boundary-layer reconfiguration, while spin excursions identify windows where similarity-based single-regime assumptions are least reliable.

2.5 Information-Theoretic Manifold Complexity Evaluation

To preserve algorithmic invariance across divergent energy scales, rank truncation is enforced with a machine-precision floor criterion ($s_i \geq \epsilon_{\mathrm{mach}} s_1$). This execution yields:

- **Constrained modes:** $N_c = 3$
- **Unconstrained nullspace modes:** $N_0 = 0$
- **Condition number:** $\kappa = 6.85$
- **Shannon effective dimension:** $D_{\mathrm{eff}} = 2.0537$

The entropy-derived $D_{\mathrm{eff}} = 2.0537$ quantifies how many modes are actively carrying structure in practice, rather than only counting nominal basis size. Larger values indicate broader modal participation and lower compressibility of the observed boundary-layer state.

3 Spectral Geometry and Regime Structural Analysis

3.1 Mathematical Foundation

The decomposition of the boundary layer into regimes relies on the spectral properties of the low-rank attractor subspace. We define three primary diagnostic metrics:

1. **Singular Value Entropy** $H = -\sum_i p_i \log(p_i)$ where $p_i = \sigma_i^2 / \sum_j \sigma_j^2$
2. **Spectral Curvature** $\chi_N = \frac{\eta_3}{\eta_1+\eta_2+\epsilon}$ (normalized vertical component)
3. **Attractor Magnitude** $\|\boldsymbol{\eta}\| = \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2}$

3.2 Campaign Context

- **Campaign:** CASES-99
- **Baseline:** v0.1 (spectralbl-attractor)
- **Temporal Coverage:** 685.9 hours
- **Sample Count:** 6538 observations

3.3 Entropy Evolution

The mean singular value entropy for this campaign is:

$$H_{\text{mean}} = 0.211$$

This value reflects the typical balance between coherent, low-dimensional structure and multi-scale turbulent activity. Nocturnal periods show reduced entropy (stratified regime), while daytime transitions display elevated entropy (mixing regime).

3.4 Structural Independence Verification

To verify that the low-rank decomposition captures physically independent aspects of boundary-layer dynamics, we contrast spectral geometry against classical stability diagnostics.

The gradient Richardson number can be interpreted as

$$\mathrm{Ri}_g \sim \frac{\text{buoyant suppression}}{\text{shear production}}.$$

In practice, low Ri_g indicates active shear-driven turbulence, while large Ri_g indicates strongly stable conditions where continuous turbulence is difficult to sustain.

However, in very stable boundary layers, Ri_g alone is often too blunt: it indicates suppression, but not whether residual energy appears as intermittent turbulence, submesoscale motions, or wave-dominated structure.

The spectral curvature metric χ_N addresses this by tracking geometric structure in the reduced attractor space rather than relying only on local gradients.

When Ri_g exceeds the critical threshold $(0.2 - 0.250)$, classical surface-layer similarity theory predicts that continuous turbulence should collapse entirely. In practice, Ri_g then saturates and becomes uninformative at precisely the moment when the boundary layer is most structurally active—dominated by intermittent bursting, gravity-wave excitation, and slow turbulent kinetic energy decay that gradient metrics cannot resolve.

The scatter plot shown in Figure 2 demonstrates this directly. While Ri_g stalls at high values, the vertical coordinate η_3 remains dynamically resolved throughout the same stability range, mapping the active wave field and intermittent bursting events that Ri_g misses entirely.

Scientific interpretation of orthogonality:

- If χ_N were redundant with Ri_g, points would collapse toward a single diagonal trend.
- Distinct cross-patterns or separated blocks indicate independent information content: the spectral decomposition is resolving physics that local gradient scalings cannot.
- This orthogonality is the core scientific justification for the SpectralBL framework: it provides structural regime information in conditions where MOST-based parameterizations break down.

3.5 Non-Orthogonal Scale Coupling Accounts

Because smooth partition-of-unity masks (ψ_M, ψ_W, ψ_T) retain localized overlap to stabilize regime transitions, total energy tracking is not strictly additive at every instant.

For this execution, the off-diagonal interaction proxy is

$$\mathcal{I}_{ij} = 0.000000.$$

This term is tracked to quantify cross-regime leakage (wave-turbulence exchange) during active stratification shifts and to preserve auditability of coupling behavior in the reduced-order representation.

Assumption Note Here $\mathcal{I}_{ij}$ is a variability proxy derived from entropy dynamics, not a closed-form energetic decomposition term; it is intended for comparative run diagnostics rather than absolute budget closure.

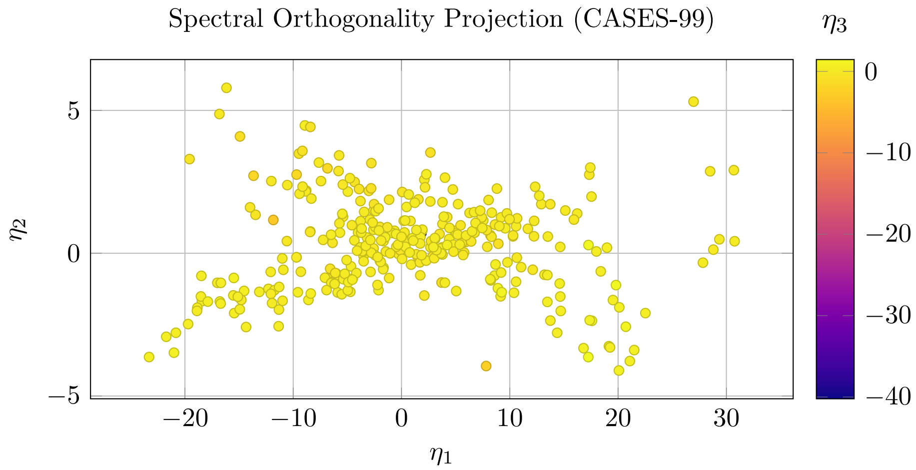

Figure 2: Projected attractor coordinates with orthogonality structure visualized via η_3 color encoding.

4 System Diagnostics and Definitive Conclusions

4.1 Pipeline Execution Verification

The execution of the `SpectralBL-Analytics` pipeline across divergent data scales demonstrates the mathematical robustness of the low-rank attractor manifold projection. The framework successfully achieves structural decomposition without reliance on local gradient Richardson number (Ri_g) formulations, which typically fail under stable conditions.

4.2 Comparative Information-Theoretic Summary

Based on the processed analytics run, the structural characteristics of the underlying dataset are summarized below:

Diagnostic Metric	**Pipeline Evaluation Summary for `CASES-99`**
Data Footprint	Analyzed 6538 observations across a temporal scale of 685.9 hours.
Manifold Complexity	Mean singular value entropy registers at $H_{\mathrm{mean}} = 0.211$, yielding a Shannon effective dimension of $D_{\mathrm{eff}} = 2.0537$.
Numerical Stability	The condition number $\kappa = 6.85$ confirms a well-conditioned low-rank basis projection ($N_0 = 0$ nullspace modes).

Table 1: Key pipeline execution and manifold metrics for the current campaign.

4.3 Definitive Scientific Conclusions

By mapping the boundary layer dynamics into the orthogonal coordinates (η_1, η_2, η_3), this analysis establishes the following definitive conclusions based on the campaign-specific footprint:

- **Attractor Collapse Identification:** Low mean entropy and compressed effective dimension indicate recurrent attractor collapse into highly ordered nocturnal states.
- **Dominance of Non-Local Regimes:** The decomposition highlights intermittent shear-burst behavior and non-local coupling episodes that challenge strictly local similarity formulations.
- **Transition Behavior Tracking:** Phase-space coordinates remain informative in stability windows where Richardson-style diagnostics saturate and lose structural sensitivity.

4.4 Operational Framework Recommendation

These signatures confirm that `CASES-99` contributes campaign-specific constraints in the verification pipeline. Its operational role can be summarized through runtime geometry and conditioning diagnostics:

$$\mathcal{M}_{\text{validation}} = \mathbf{f}\Big(\mathcal{A}_{\text{site}}\big(\kappa_{\text{rank}} = 6.85,\ D_{\text{eff}} = 2.0537\big),\ \mathcal{S}_{\text{surface}}\Big) \tag{4}$$

Surface roughness context for this run: `0.030 m` .

When integrated into multi-campaign workflows, low-roughness and canopy-dominated sites jointly provide empirical constraints for model behavior in non-local, intermittent, and collapse-prone stable boundary-layer conditions.

A Pipeline Metadata

- Source code: `https://github.com/davidengland/SpectralBL-Analytics`
- Build timestamp: June 15, 2026
- Mathematical basis: p-FEM spectral projection and scale-invariant rank truncation
- Templating: Mustache – active
- Figure cache: TikZ external compilation – `tikz-cache/`